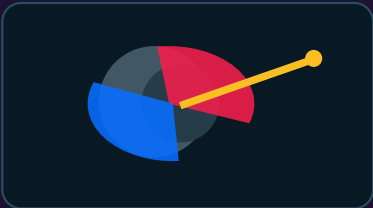


COLOR DOPPLER vs SPECTRAL DOPPLER

COLOR DOPPLER

MAP

WHERE is blood flowing?



- Localizes flow within 2D anatomy
- Shows map-relative direction
- Finds jets, shunts, regurgitation
- Flags turbulence or aliasing

Common targets: septa, AV/semilunar valves, outflows, arches, venous pathways

SPECTRAL DOPPLER

MEASUREMENT

HOW is blood moving?



- Measures peak velocity
- Shows direction versus baseline
- Displays waveform shape over time
- Supports rhythm and AV timing

Common targets: AV inflows, aortic/pulmonary outflows, DV, arch, mechanical AV interval

Clinical question	COLOR	SPECTRAL
Where is flow?	Primary tool	Not primarily
Direction?	Map-dependent	Versus baseline
Regurgitation?	Detect + localize	Velocity + timing
Peak velocity?	Not primary	Yes
Pattern over time / AV interval?	No waveform	Yes

VSD

- C: Flow crossing septum
- S: Velocity/timing if needed

AORTIC STENOSIS

- C: Aliasing/turbulence
- S: Peak systolic velocity

READ THE DISPLAY CORRECTLY

Color direction depends on the displayed color map and transducer reference. Red/blue does **not** mean oxygenated/deoxygenated.

TRICUSPID REGURGITATION

Color: localizes the jet into the RA.
Spectral: characterizes velocity and timing when indicated.

NO WAVEFORM = NO 76828

COLOR ALONE DOES NOT SUPPORT 76828

93325: color-flow mapping, stored images/clips.
76828: actual spectral interrogation with stored PW/CW waveforms.

Five-second rule: COLOR = WHERE | SPECTRAL = HOW

Educational aid. Verify current CPT guidance, payer policy, and documentation requirements.